Tab 1

Lab Session 1

1. Greeting Program

Write a program that asks the user for their name and age, then prints:

Hello [name], you are [age] years old!

2. Simple Calculator

Ask the user for two numbers. Print the sum of those numbers using ***print()***.

3. Favorite Things

Take 3 inputs: favorite color, favorite food, and favorite hobby.

Print a sentence like: ***My favorite color is blue, I love pizza, and I enjoy reading.***

4. Area of a Rectangle

Ask the user for the length and width of a rectangle.

Calculate and print the area with a message: ***The area is [result]***.

5. Name Length

Ask the user for their full name.

Print a message showing how many characters are in their name.

Hint: Use ***len()*** function.

6. Temperature Converter

Ask the user for temperature in Celsius.

Convert it to Fahrenheit using the formula ***F = C 9/5 + 32*** and print the result.

7. Mad Libs Game

Ask for a noun, verb, and adjective.

Print a funny sentence like: ***The [adjective] [noun] likes to [verb] all day.***

8. Age in Months

Ask the user for their age in years.

Print their approximate age in months.

Hint: ***1 year = 12 months***.

9. Personalized Welcome

Ask for the user's first and last name separately.

Print a welcome message in 2 lines:

Welcome, [First Name]!

Your full name is [First Name] [Last Name].

10. Mini Bio Generator

Ask the user for: name, city, and favorite animal.

Print a short bio in one line:

Hi, I'm [name] from [city]. My favorite animal is [animal].

Tab 2

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Lab Session 2

Task 1: The Classic Welcome Banner

Objective: Use string multiplication to create a dynamic, scalable border around a welcome message.

```
==========================================
=  WELCOME TO PYTHON PROGRAMMING LAB  =
==========================================
```

Task 2: Digital Receipt Generator

Objective: Use f-string alignment specifiers (< for left, > for right) and dashes to format a clean retail receipt.

```
----------------------------------------------------------
|          RETAIL RECEIPT                       |
----------------------------------------------------------
| Wireless Mouse                 $   25.50  |
| Mechanical Keyboard         $   85.00  |
| USB-C Cable                    $   12.75  |
----------------------------------------------------------
| TOTAL:                          $  123.25 |
----------------------------------------------------------
```

Task 3: Multi-Format Number Matrix

Objective: Display a single integer in decimal, binary, octal, and hexadecimal formats using f-string type codes inside a grid.

```
+--------------------+-------------------+---------+
|  DECIMAL        |   BINARY        |    HEX |
+--------------------+-------------------+---------+
|    156             |  10011100      |     9C   |
+--------------------+-------------------+---------+
```

Task 4: Temperature Conversion Table

Objective: Calculate Celsius to Fahrenheit conversions and display them using a clean, tabular border structure.

```
╔══════════════╦════════════════╗
║ Celsius (°C) ║ Fahrenheit (°F) ║
╠══════════════╬════════════════╣
║     0.0      ║      32.0      ║
║    10.0      ║      50.0      ║
║    20.0      ║      68.0      ║
║    30.0      ║      86.0      ║
║    40.0      ║     104.0      ║
╚══════════════╩════════════════╝
```

Task 5: Dynamic Multiplication Grid

Objective: Build a mini math matrix utilizing uniform widths to keep custom row/column dividers perfectly aligned.

```
╔═════════════════════╗
║  MULTIPLICATION: 7  ║
╠══════════╦══════════╣
║  7 x 1   ║   7      ║
║----------+----------║
║  7 x 2   ║   14     ║
║----------+----------║
║  7 x 3   ║   21     ║
║----------+----------║
║  7 x 4   ║   28     ║
║----------+----------║
║  7 x 5   ║   35     ║
╚══════════╩══════════╝
```